Efficient Coding of Drifting Images or maybe Statistics of Moving Images

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Keywords: Manuscript, journal, instructions

Abstract

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1 Introduction

- Eyes are never still. Saccades allow for looking at things without moving head. But even when head is still and eyes are fixating, movements persist.
- Such fixational movements have been shown to be a feature, not a bug.
- https://www.pnas.org/doi/pdf/10.1073/pnas.1412059112 show that fixational eye movements contribute produce more decorrelated RGC outputs.
- https://www.nature.com/articles/s41467-024-52304-7.pdf show that fixational drift enhances precision of neural representations. Ideal observer model on RGC outputs produced from simulated drift inputs.

- Ratham and Roorda paper shows this psychophysically, and models have been developed to explain results (alex anderson and olshausen).
- Fixational drift isnt restricted to vertabrites. Insects also exhibit fixational drift (Fenk,.., Maimon). A steady eye is becoming the exception, not the rule.
- Meanwhile, circularly symmetric center-surround receptive fields are obiquitous in eraly visual systems.
- Natural scenes are highly correlated. Leading theory is that center surround receptive fields are bandpass in spatial freqeuncy and serve as whitening filters such that the output is decorrelated. Decorrleated representatoin is better as it has reduced reduancy (pair-wise statistiscs-wise) (barlow) thus neural systems forming a more energy efficient code (not represeting redundancies).
- Kuang,..., Rucci reporte whiting effect of spatiotemporal powerspectrum. If signal is already whitened by fixational eyemovements, what are center-surround receptive fields doing?

How we address this:

- Derive analytical form of fixational drifting natual images. Present this information clearly and in one place.
- Show that it is parially whitened.
- Derive receptive fields that maximize information transfer subject to input and output noise, and contraints on output power and temporal response.
- Demonstrate that optimal RFs are spatially bandpass, and amplify singal in the part of the power spectrum that hasnt been whitened.
- Also optimize over drift diffusion constants and show that under realistic amounts of noise the optimal diffuision is consistent with previously measured values.
- Predictions about Diffusion as a function of noise, and show that, under fixed levels of diffusion, increased noise attenuates surround inhibition consistent with [1].

Goal is to understand the statistical structure of images formed on moving sensors (such as animal eyes) and understand it's implications on retinal coding. Deriving a complete statistical description of images formed in moving optical systems is difficult. We make the simplifying assumption that the statistics of the movement changes in stationary increments and independent of the scene being imaged and analyze pairwise statistics. Under these assumptions, we can derive an analytical form of the autocorrelation function $C_z(\xi, \tau)$, which can then be studied in the Fourier domain. The power spectrum is generally not separable in spatial and temporal frequencies. We study the structure of the power spectrum arising from power-law natural images undergoing Brownian motion. Assuming well-known statistical structure of natural images, the resulting power spectrum has interesting properties. Specifically, slicing at a spatial or

temporal frequency, power is flat up to a cut-off frequency, and then falls as a function of f^{-4} or ω^{-2}. Derive efficient coding models that predict optimal filters that are center-surround.

Statistical structure of drifting natural images (either under empirically observed drifting eye motion or Brownian) has been reported before (Rucci ....). Here, we seek describe the statistical structure of the data, and understand its implications for retinal coding.

2 Correlation structure of moving images

For clarity, we present the derivations in one spatial dimension. The extension to two spatial dimensions is straightforward. Let $s(x)$ denote a static image, where x denotes position. The time-varying displacement $u(t)$ produces the movie

$$z(x,t) = s\big(x - u(t)\big). \tag{1}$$

We assume that $u(t)$ has stationary displacement increments and is independent of the static image $s(x)$. Under the assumptions of $u(t)$ above, the probability density of displacements is a function of spacial displacement ξ and temporal lag τ, given by

$$u(t+\tau) - u(t) \sim \rho(\xi, \tau). \tag{2}$$

The space-time autocorrelation $C_z(\xi,\tau)$ can be written as a simple spatial convolution of the spacial autocorrelation function $C_s(\xi)$ and the movement density function $\rho(\xi,\tau)$,

$$C_z(\xi,\tau) = \big\langle z(x,t)\, z(x+\xi, t+\tau)\big\rangle_{x,t} \tag{3}$$

$$= \big(\rho(\cdot,\tau) * C_s\big)(\xi) \tag{4}$$

The derivation is given in Appendix A.1. Since the spatial convolution becomes multiplication in the Fourier domain, the space-time power spectrum is

$$C_z(k,\omega) = \rho(k,\omega)\, C_s(k). \tag{5}$$

- multiplication relies on fact that image is not changing in time (i.e., $s(x,t)$).
- Movement – all its contributions are given by rho – acts like a filter on static spectrm Cs. This filter pushes spatial information into time.
- This Spatiotemporal power spectrum is not separable. as rho depends on both. this is in contrast to assumptions made by Ocko, Jun, etc.

2.1 Brownian drift and partial whitening

To describe fixational eye drift, we specialize the motion model to Brownian motion.

Let $u(t)$ evolve stochastically from initial position x_0 via Brownian motion W_t with diffusion constant D:

$$du(t) = \sqrt{2D}dW(t), \quad u(0) = x_0. \tag{6}$$

The probability distribution of the SDE solution is $\rho(\xi, \tau)$, where ξ and τ is space and time, respectively

$$\rho(\xi, \tau) = \frac{1}{\sqrt{4\pi D|\tau|}} \exp\left(-\frac{\xi^2}{4D|\tau|}\right). \tag{7}$$

The space-time fourier transform of $\rho(\xi, \tau)$ has a simple form (see appendix A.2 for derivation details). A well known property of static natual images is that the power spectrum is power-law in spatial frequency. This means that $C_s(k) \propto k^{-\beta}$ for some $\beta > 0$. For Brownian drift, the stimulus power spectrum depends on the diffusion constant D. We make this dependence explicit by writing $C_z(k, \omega; D)$. The resulting spectrum is given by the static-image power spectrum $C_s(k)$ multiplied by the space-time Fourier transform of ρ:

$$C_z(k, \omega; D) = \rho(k, \omega)\, C_s(k) \tag{8}$$

$$= \frac{2Dk^{2-\beta}}{(Dk^2)^2 + \omega^2} \tag{9}$$

for $D > 0$.

For the natural-image exponent $\beta = 2$, the spatial-frequency dependence in the numerator of Eq. 9 cancels. At a fixed nonzero temporal frequency, the spectrum is approximately flat below the spatial crossover $k = \sqrt{|\omega|/D}$ and falls as k^{-4} above it. Likewise, at fixed spatial frequency, the temporal spectrum is approximately flat below $|\omega| = Dk^2$ and falls as ω^{-2} above it. Figure 1 shows this partial whitening and the dependence of the spectrum on the static-image exponent and diffusion constant.

The fall of the power spectrum differs from previous reports of natural-movie statistics. Dong and Atick [2] analyzed handycam footage they recorded as well as Hollywood movies and report that, in general, the spatiotemporal power spectrum is inseperable, but can be approximated as seperable under certain conditions. Spatial powerlaw structure in Hollywood movies also been reported by [3]. Fixed position camera [4] has recorded power law spatial frequency fall off. (also see [5]). Work within the efficient coding literature has derived theory of retinal function from spatiotemporal powerspectrums that are both powerlaw in space and time and seperable [5, 6]. The powerspectrum of drifting natural images is neither powerlaw in spatial and temporal frequency, nor is it seperable.

Decorrelation of natural-image inputs is a classic explanation for center–surround receptive fields. Brownian drift already flattens the stimulus spectrum over part of the spatiotemporal frequency range. This raises a question for efficient coding: if eye movements reduce spatial correlations before neural processing, why should subsequent processing still favor center–surround filters?

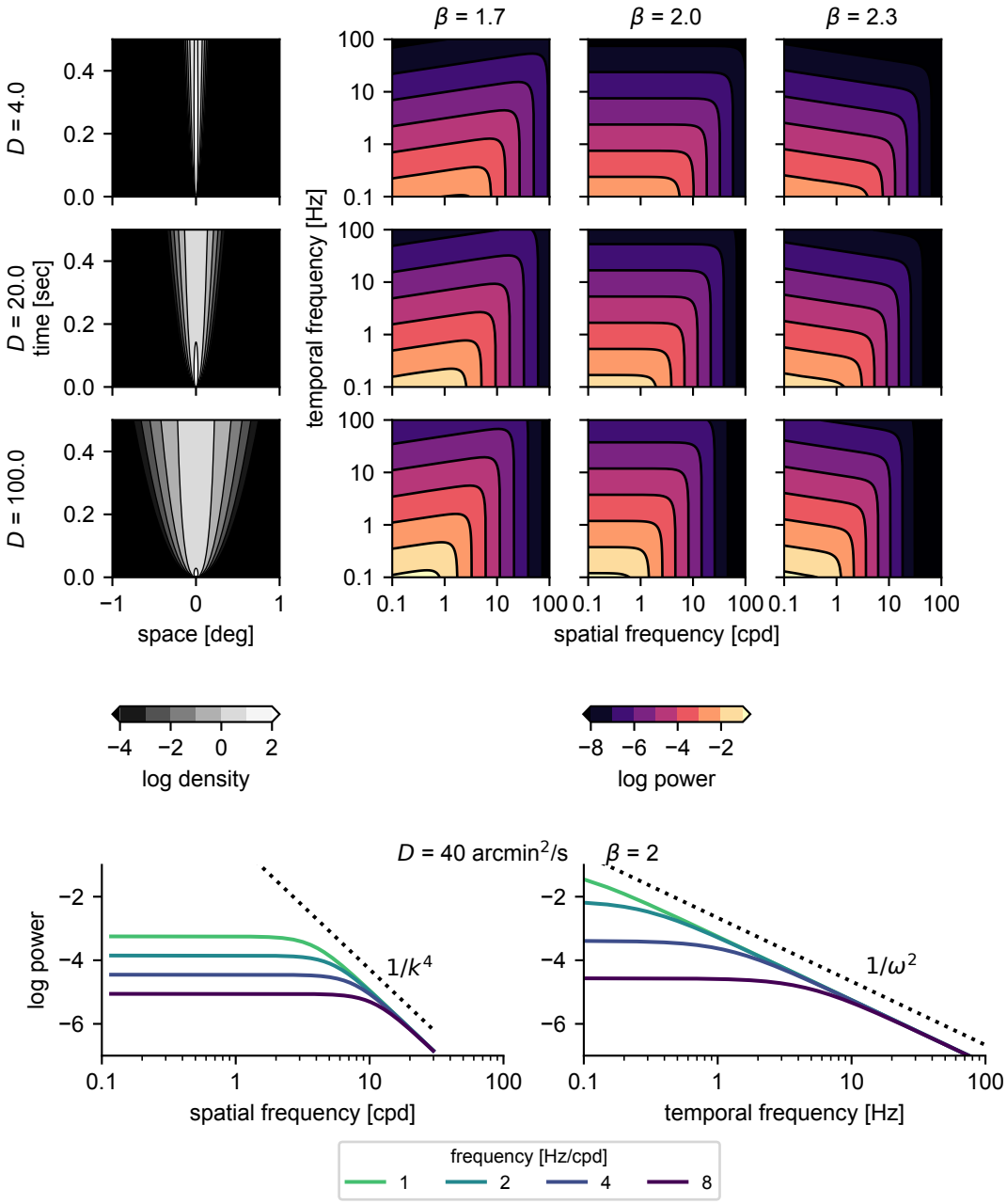


Figure 1: Partial whitening of drifting natural images. Top: spatiotemporal power spectra for different static-image exponents and diffusion constants. Bottom: spatial and temporal cross-sections for $\beta = 2$ and $D = 40$ arcmin^2/s, with reference slopes k^{-4} and ω^{-2}.

3 Efficient coding of drifting images

Brownian drift flattens the stimulus spectrum over part of the spatiotemporal frequency range. This raises a question for efficient coding: if eye movements already reduce spatial correlations, why should subsequent neural processing favor center–surround receptive fields? To address this question, we optimize a linear encoding model for the statistics of drifting images.

To address this question, we model the neural response using a linear space-time filter w. The response $r(x,t)$ of the linear model is a function of the input $z(x,t)$ and noise $n_{in}(x,t)$ and output noise $n_{out}(x,t)$,

$$y(x,t) = \big(w * (z + n_{in})\big)(x,t), \quad r(x,t) = y(x,t) + n_{out}(x,t). \tag{10}$$

Here, $*$ denotes convolution over space and time. We take the input and output noise to be independnt Gaussian noises with zero mean, with variance σ_{in}^2 and σ_{out}^2, respectively. We seek to find the filter w and drift constant D that maximizes the mutual information between the input z and response r, subject to a power and temporal derivative constraint on the response.

Constraints on the filter are necessary because, without them, the filter gain can be trivially increased to surpass the output noise [7]. To prevent this, we constrain the output variance of the filter, similar to prior work [1, 8]. We introduce an additional constraint on the temporal derivative of the outputs as high frequency responses to not come for free in neural systems, they require larger neurons and increased metabolic cost. This constrainst is also applied to filter outputs before output noise is added. The information maximization objective together with the variance and smoothness constraints results in the follwing optimization problem

$$\begin{aligned} \underset{w,D}{\text{maximize}} \quad & I(Z;R) \\ \text{subject to} \quad & \left\langle y(x,t)^2 \right\rangle_{x,t} + \eta \left\langle \dot{y}(x,t)^2 \right\rangle_{x,t} = 1. \end{aligned} \tag{11}$$

where $\eta \geq 0$ controls the strength of the temporal smoothness penalty. The procedure for solving this optimization problem is described in full in Appendix A.3. In brief, we express the objective and constraint in the Fourier domain and introduce a Lagrange multiplier. In doing so, we obtain a closed-form expression for the optimal filter power at each fixed diffusion constant D. We determine the multiplier numerically so that the combined variance and smoothness constraint is satisfied. We then find the optimal diffusion constant D^* by searching over D, reoptimizing the filter at every trial value.

We first build intuition using static natural images, then examine the optimal filter for drifting images at a fixed diffusion constant, and finally allow diffusion to adapt as the noise levels change. The effect of the temporal penalty on the optimal diffusion constant is shown in Fig. S1; we hold η fixed for the noise comparisons below.

3.1 Center–surround filters despite partial whitening

Before applying this model to drifting images, we consider the illustrative example of the whitening or decorrleting theory of the retina for the case of static natural images [1, 9, 10]. Its well known that the spatial power spectrum of natural images is

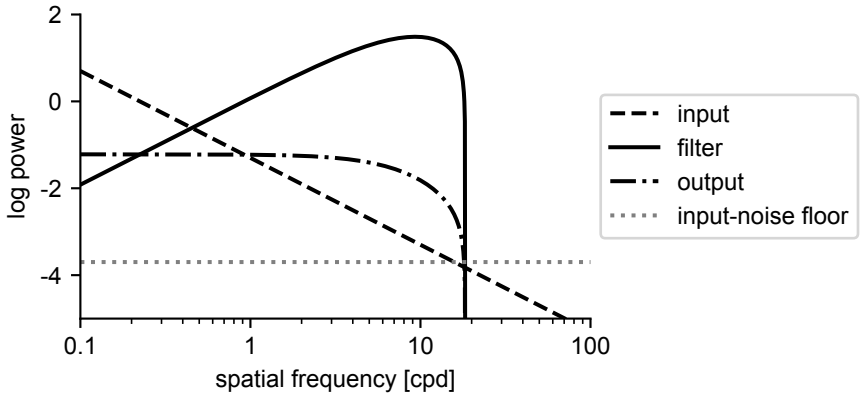


Figure 2: Noise-limited whitening of static natural images. Dashed: input signal power $C_s(k) \propto k^{-2}$. Solid: optimal filter power gain $|w(k)|^2$. Dash-dotted: filtered signal power $C_s(k)|w(k)|^2$, approximately flat at low frequencies and attenuated at high frequencies. Dotted: input-noise power. The filter maximizes information subject to a unit variance constraint before output noise is added, without a temporal penalty. The output curve excludes noise; filter power gain and signal power have different units.

approximately $C_s(k) \propto k^{-2}$ [11]. A filter with power gain $|w(k)|^2 \propto k^2$ would flatten, or whiten, the signal spectrum, since the filtered signal has power $C_s(k)|w(k)|^2$. With input noise and a limited response-power budget, however, amplifying weak high-frequency components also amplifies noise. The optimal filter therefore approximately whitens the signal at low spatial frequencies and suppresses sufficiently high frequencies, with the cutoff determined by both noise levels and the power constraint.

This provides intuition for center–surround filtering. Opposing contributions from the center and surround attenuate slowly varying inputs, while the finite spatial extent of the center limits sensitivity to high frequencies. The resulting bandpass response is consistent with noise-limited whitening.

We now apply the same approach to drifting images, including the temporal smoothness constraint. We fix the noise at $\sigma_{in} = 0.2$ and $\sigma_{out} = 20$, similar to prior work [5, 12–15]. Set $\eta = 0.003$ which results in $D^* = 40$ arcmin^2/s, the optimal filter is spatially bandpass for the illustrated noise conditions (Fig. 3). The filter preferentially amplifies a range of spatial frequencies beyond the spectral crossover, where the stimulus power still falls as k^{-4}, while suppressing frequencies at which noise dominates. Using a spatially symmetric, temporally minimum-phase reconstruction (Appendix A.4), the corresponding spatial kernel has a central lobe flanked by antagonistic surrounds.

This resolves the apparent conflict: drift whitens only part of the spectrum. The remaining correlations, together with noise and the output constraint, can still favor center–surround filtering. The whitening produced by movement therefore does not eliminate the role of bandpass neural filtering for optimal coding of drifting natural images.

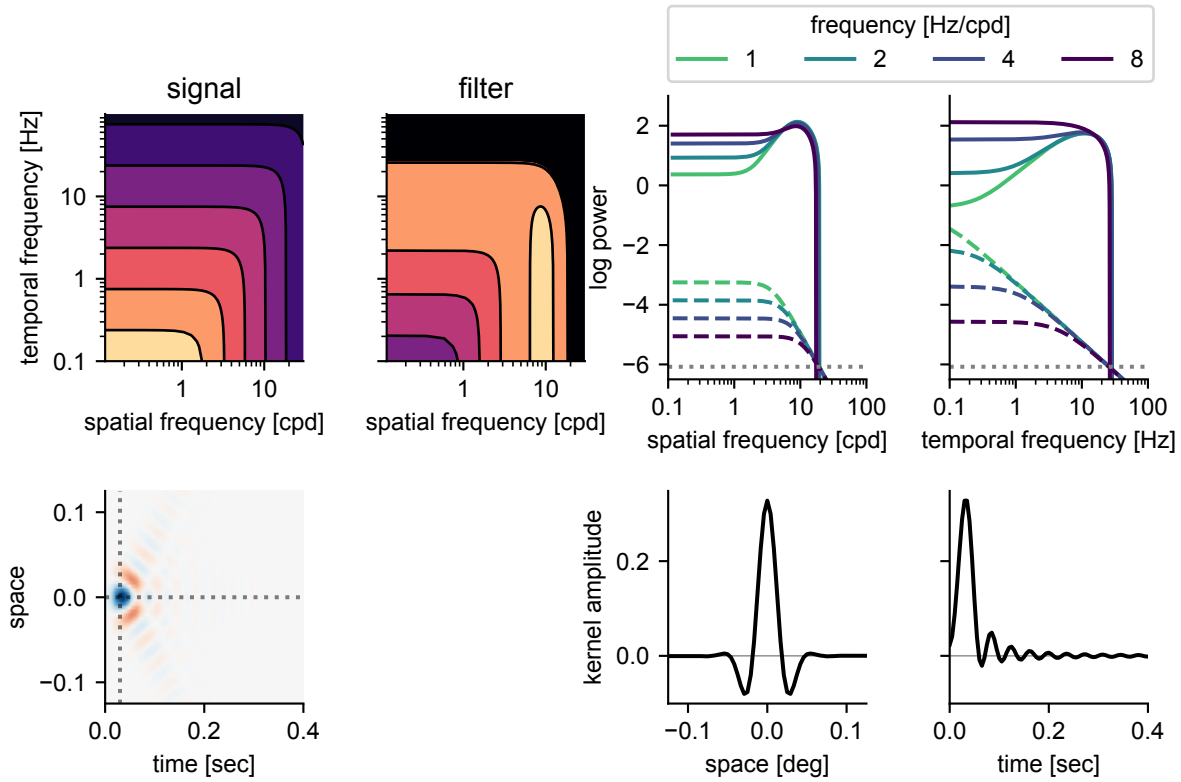


Figure 3: Efficient coding preserves center–surround filtering after Brownian drift. Top: the stimulus and optimal filter power spectra, with spatial and temporal cross-sections; solid curves show filter power and dashed curves show stimulus power. Bottom: the reconstructed space-time kernel and its spatial and temporal cross-sections through the peak response. The spatial cross-section is taken at the time of peak response at the kernel center; the temporal cross-section is taken at the kernel center ($x = 0$). Diffusion is fixed at $D^* = 40$ arcmin^2/s

3.2 Joint adaptation of filters and drift

Having established that center–surround filters remain favored at fixed drift, we next ask how the optimal diffusion constant and filters change with noise. For each pair of input and output noise levels, we search over D and recompute the optimal filter subject to the same variance and smoothness constraint. The information rate has a maximum at a finite diffusion constant for the illustrated conditions (Fig. 4, top left). Increasing input noise shifts this optimum toward larger diffusion constants, whereas increasing output noise shifts it toward smaller values (Fig. 4, top right).

Allowing diffusion to adapt also changes the predicted response of the kernels to input noise. In the adaptive case, the spatial kernels broaden while retaining antagonistic surrounds, and the temporal kernels peak earlier as input noise increases (Fig. 4, bottom left). With diffusion fixed, the surround weakens as input noise increases over the range shown (Fig. 4, bottom right). These comparisons use the noise levels indicated in their respective legends. Joint adaptation of movement and filtering can therefore produce a different pattern of receptive-field changes from adaptation of the filter alone.

Optimal diffusion constants increase rapidly as a function of input noise, quickly exceeding a threshold that would be physically possible to move an eye. We haven't considered such contraints here. Weird and interesting prediction though that drift could increase information acquisition in low contrast or high noise conditions.

Discussion

other notes

- https://www.jneurosci.org/content/jneuro/34/38/12701.full.pdfm – aytekin victor rucci 2014 – The results of our study show that, during normal fixation, the spatiotemporal signals impinging on retinal receptors are already whitened within the range of temporal frequencies at which ganglion cells respond best. Whats our response? This whitening occurs before any neural processing takes place. "Thus, neural decorrelation, is not a viable theory to explain the filtering characteristics of receptive fields. contrast sensitivity functions of retinal ganglion cells further enhance high spatial frequencies above and beyond what is required for redundancy removal, leading to an enhancement of the luminance discontinuities in the scene. Critically, this edge enhancement happens in the temporal domain: it occurs in the synchronous firing of retinal ganglion cells, yielding a temporal code that takes advantage of the notion that synchronous responses tend to propagate more reliably than asynchronous ones."

- https://www.pnas.org/doi/pdf/10.1073/pnas.1412059112 Decorrlation of retinal response to natural scenes by fixational eyemovements

- Main contributions: Describing pair-wise statistical structure of stochastically moving images. Focus in on drift, which produces a whitening effect. Despite this apparent structure, we show that circularly symmetric center-surround receptive fields of RGCs are predicted (optimal) under efficient coding objective. Gained

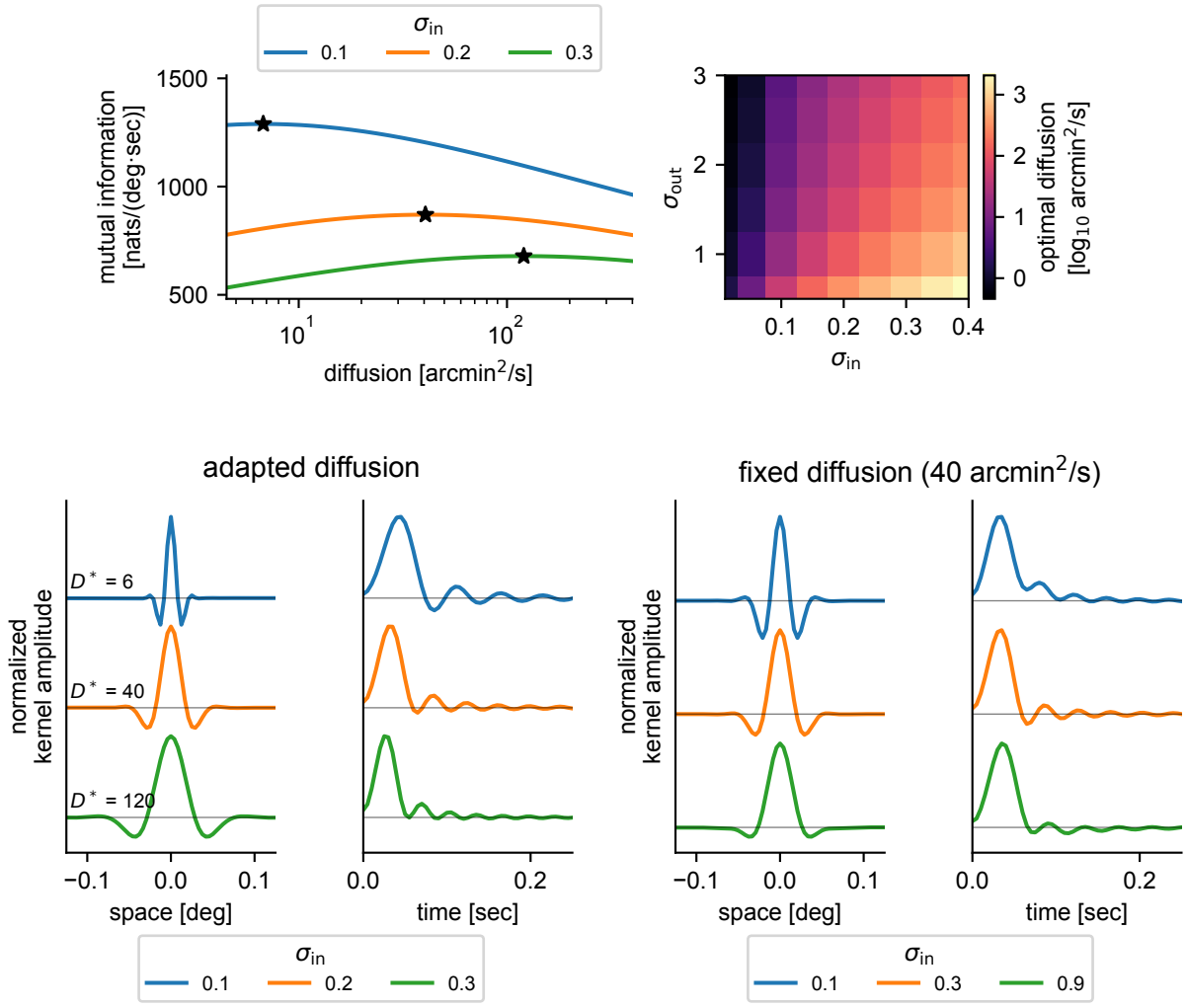


Figure 4: Noise-dependent adaptation of drift and filtering. Top left: information rate as a function of diffusion at three input noise levels; stars mark the optimized diffusion constants. Top right: optimal diffusion as a function of input and output noise. Bottom: normalized spatial and temporal cross-sections through the peak response of each space-time kernel, with diffusion optimized separately at each input noise level (left), or fixed at 40 arcmin^2/s (right). Spatial cross-sections are taken at the time of peak response at the kernel center; temporal cross-sections are taken at the kernel center ($x = 0$). Kernel traces are vertically offset for clarity. The two kernel comparisons use different input-noise ranges, as indicated in their legends.

understanding in linear model; nonlinear model is consistent. The reconciliation relies on the specific structure of the power spectrum. spatial Power drops as k^{-4} after a cut-off frequency. Still signal left to amplify, thus optimal filters become band-pass in space.

- Generate new prediction: higher noise (low contrast) eyes may drift more? Drifting allows for reshaping statistics before any neural processing occurs.
- Relevant work – rucci.
- Earlier reports of spatio temporal statistics of natural images – Dong and Atick, Eckea M P and Buchsbaum G 1993,
- Efficient coding literature – Jun, Ocko.
- Nonseparable power spectrum. Receptive fields reported to be separable in space time. Maybe due to receptive field measurement procedure. Nonseperable receptive fields could be produced from temporal dynamics of center and surround mechanisms in retina.
- Nonstationarity – saccades.

A Appendix

A.1 Autocorrelation of moving images

Follow derivation process of Dong and Atick 1995 [2]. That is, get an equation that describes the spatio-temporal correlations $C_z(\xi, \tau)$ as a function of space ξ and time τ lags. Then we can take the fourier transform in space and time. Define spatial-temporal autocorrelation function

$$C_z(\xi, \tau) = \mathbb{E}_{x,t}\Big[z(x,t)\, z(x+\xi, t+\tau)\Big] \tag{12}$$

$$= \mathbb{E}_{x,t}\Big[s\big(x - u(t)\big)\, s\big(x + \xi - u(t+\tau)\big)\Big]. \tag{13}$$

We assume that $u(t)$ is governed by a stochastic process. For a temporal lag τ, define the spatial displacement

$$\Delta u(\tau) := u(t+\tau) - u(t), \tag{14}$$

where the density of this displacement is given by $\Delta u(\tau) \sim \rho(\xi, \tau)$ and ξ denotes a possible displacement value. Let $l = x - u(t)$ and substitute the stimulus $z(x,t)$ above with $s(x - u(t))$:

$$C_z(\xi, \tau) = \mathbb{E}_{x,t}\Big[s\big(x - u(t)\big)\, s\big(x + \xi - u(t+\tau)\big)\Big] \tag{15}$$

$$= \mathbb{E}_{l,t}\Big[s(l)\, s\big(l + \xi - \Delta u(\tau)\big)\Big] \tag{16}$$

Define the static image autocorrelation

$$C_s(\xi) := \mathbb{E}_x[s(x)s(x+\xi)]. \tag{17}$$

Assuming translation invariance, we can take expectation over brownian paths which then describes the spatiotemporal correlation, which can be equivalently described as spatial convolution

$$C_z(\xi, \tau) = \mathbb{E}_{\Delta u(\tau)}\Big[C_s\big(\xi - \Delta u(\tau)\big)\Big] \tag{18}$$

$$= \int_{-\infty}^{\infty} C_s(\xi - \delta)\, \rho(\delta, \tau)\, d\delta \tag{19}$$

$$= (C_s * \rho(\cdot, \tau))(\xi). \tag{20}$$

A.2 Power spectrum under Brownian drift

$$\rho(\xi, \tau) = \frac{1}{\sqrt{4\pi D|\tau|}} \exp\left(-\frac{\xi^2}{4D|\tau|}\right). \tag{21}$$

The spatial Fourier transform of Gaussian ρ is another Gaussian:

$$\rho(k, \tau) = e^{-D|\tau|k^2}. \tag{22}$$

Putting these two together, you get

$$C_z(k,\tau;D) = C_s(k)\, e^{-D|\tau|k^2} \tag{23}$$

Now take the Fourier transform in time τ:

$$C_z(k,\omega;D) = \int_{-\infty}^{\infty} C_z(k,\tau;D) e^{-i\omega\tau}\, d\tau \tag{24}$$

$$= C_s(k) \int_{-\infty}^{\infty} e^{-Dk^2|\tau|} e^{-i\omega\tau}\, d\tau. \tag{25}$$

The integral can be solved by splitting into two integrals from $(-\infty, 0)$ and $[0, \infty)$ to deal with the $|t|$ in exponent which ends up working out into a simple form

$$\int_{-\infty}^{\infty} e^{-a|t|} e^{-i\omega t}\, dt = \frac{2a}{a^2+\omega^2} \tag{26}$$

with $a = Dk^2$, we get

$$C_z(k,\omega;D) = C_s(k)\, \frac{2Dk^2}{(Dk^2)^2+\omega^2} \tag{27}$$

up to a constant. Note that in the limit as D goes to zero, all power lies at $\omega = 0$, and within this plane the power is the input static image power

$$\lim_{D\to 0^+} C_z(k,\omega;D) \propto C_s(k)\, \delta(\omega) \tag{28}$$

A.3 Optimization of linear filters and diffusion

The squared magnitudes of the optimal linear filters $v(k,\omega) = |w(k,\omega)|^2$ can be solved for analytically in the Fourier domain. The spatiotemporal response of the linear model is given by

$$r(k,\omega) = w(k,\omega)\big(z(k,\omega) + n_{in}(k,\omega)\big) + n_{out}(k,\omega). \tag{29}$$

We distinguish the space-time noise variances σ_{in}^2 and σ_{out}^2 from their power spectral densities. For Gaussain white noise, these densities are constant over the modeled frequency band and we denote their levels by N_{in} and N_{out}. Under the Gaussian input approximation, the mutual information rate per unit space between the input z and response r is

$$I(Z;R) = \frac{1}{2} \int \frac{dk, d\omega}{(2\pi)^2}\, \log\left(1 + \frac{v(k,\omega), C_z(k,\omega;D)}{N_{in} v(k,\omega) + N_{out}}\right). \tag{30}$$

The term in the integral has the familiar form of $\log\big(1 + \text{SNR}(k,\omega)\big)$, which is then integrated over spatial and temporal frequencies.

The combined power and temporal derivative constraint can also be written in terms of spatial and temporal frequencies as a function power spectrum of the filter response $C_y(k,\omega)$,

$$\int \frac{dk, d\omega}{(2\pi)^2} \left(1 + \eta\omega^2\right) C_y(k,\omega) = 1, \quad C_y(k,\omega) = v(k,\omega)\left(C_z(k,\omega;D) + N_{in}\right). \tag{31}$$

where ω^2 arises from the temporal derivative constraint.

The information optimization objective subject to this constraint can be rewritten using a Lagrange multiplier λ. The Lagrangian for a given spatial and temporal frequency is

$$\mathcal{L}_{k,\omega} = \frac{1}{2}\log\left(1 + \frac{v(k,\omega)C_z(k,\omega;D)}{N_{in}v(k,\omega) + N_{out}}\right) - \lambda(1+\eta\omega^2)v(k,\omega)\big(C_z(k,\omega;D) + N_{in}\big). \tag{32}$$

Setting $\partial\mathcal{L}k,\omega/\partial v = 0$ at an active mode gives

$$\begin{aligned}\frac{N_{out}C_z(k,\omega;D)}{2\big(N_{out} + (C_z(k,\omega;D) + N_{in})v(k,\omega)\big)\big(N_{out} + N_{in}v(k,\omega)\big)} \\ = \lambda(1+\eta\omega^2)\big(C_z(k,\omega;D) + N_{in}\big).\end{aligned} \tag{33}$$

For positive input and output noise levels, solving for $v(k,\omega) = |w(k,\omega)|^2$ yields

$$\begin{aligned}v(k,\omega) = \frac{N_{out}}{N_{in}}\Bigg[\frac{1}{2}\frac{C_z(k,\omega;D)}{C_z(k,\omega;D) + N_{in}} \\ \times\left(\sqrt{1 + \frac{N_{in}}{N_{out}}\frac{2}{\lambda(1+\eta\omega^2)C_z(k,\omega;D)}} + 1\right) - 1\Bigg]_+ .\end{aligned} \tag{34}$$

The nonlinearity $[\cdot]_+ = \max(\cdot, 0)$ ensures that filter powers $v(k,\omega) = |w(k,\omega)|^2$ are nonnegative. For each fixed diffusion constant D, we determine the Lagrange multiplier λ numerically so that the optimal filter satisfies the combined variance and smoothness constraint in Eq. 31.

We find the optimal diffusion constant D^* by numerically searching over D to maximize the information rate, with the noise levels and η held fixed. At each trial value of D, we update the stimulus spectrum $C_z(k,\omega;D)$ and recompute the optimal filter and its Lagrange multiplier so that the constraint remains satisfied.

At the reference noise levels, increasing the temporal penalty shifts the optimum toward smaller diffusion constants (Fig. S1). This dependence shows how the temporal constraint sets the preferred drift scale; its strength is held fixed when comparing different noise conditions in the main text.

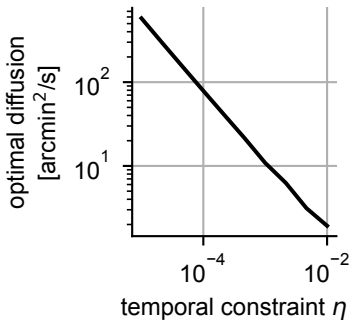


Figure S1: Dependence of the optimal diffusion constant on the temporal smoothness penalty η, with input and output noise levels held fixed. The filter is reoptimized at each trial diffusion constant.

A.4 Reconstruction of space-time kernels

We reconstruct the space–time kernel $w(x,t)$ by inverse Fourier transforming the optimal filter $w(k,\omega) = \sqrt{v(k,\omega)}\, e^{i\theta(k,\omega)}$:

$$w(x,t) = \int \frac{dk, d\omega}{(2\pi)^2}\, w(k,\omega)\, e^{i(kx+\omega t)}.$$

The optimization determines the filter amplitude but leaves its phase $\theta(k,\omega)$ unspecified. We choose a spatially symmetric, emporally minimum-phase response. Since $v(k,\omega)$ is even in k, spatial symmetry is preserved by assigning the same phase to positive and negative spatial frequencies. The minimum temporal phase is given by the negative Hilbert transform of the log amplitude $\theta(k,\omega) = -\mathcal{H}_\omega \left\{ \log \sqrt{v(k,\omega)} \right\}$.

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